FMP · MODULE 4 · WORKBOOK

PRACTICAL EVIDENCE · EQUITY SECURITIES

Workbook, answered.

Model answers to every question in the Module 4 workbook — all **10 questions, 150 marks** — each sized to its mark allocation and grounded in the Module 4 material. Use these to check your understanding and structure your own submission.

Note: these are study/model answers to learn from. The workbook is assessed as **your own work** (pass mark **65%**) — write your final answers in your own words.

MARK ALLOCATION AT A GLANCE

QUESTION	TOPIC	MARKS
Q1	Equity fundamentals & capital structure	15
Q2	Equity income, share types, listed/unlisted, DRs	15
Q3	Determinants of optimal capital structure	15
Q4	Debt versus equity	10
Q5	Primary & secondary markets, IPOs, listing	30
Q6	Listing reasons, methods, rights issue, risks	15
Q7	Gordon Growth model — shortcomings	5
Q8	Market capitalisation (FSR vs Nedbank)	10
Q9	Valuation techniques by sector	10
Q10	Dual-class shareholding case study	25
Total		**150**

HOW TO USE Match each answer's depth to its marks — a **(3)** wants three distinct points, a **(2)** wants two, a **(5)** wants a fuller paragraph or five points. Lead with the key term, then explain.

QUESTION 1 · EQUITY FUNDAMENTALS & CAPITAL STRUCTURE [15]

1.1 Main advantages of raising equity capital for a company? (3)

- **No repayment obligation** — equity is permanent capital with no fixed interest burden.
- **Lower financial risk & stronger balance sheet** — less gearing → more resilient and more creditworthy for future borrowing.
- **Shares risk & widens access** — investors share the risk; issuing/listing raises profile and the investor base, with no collateral needed.

1.2 Factors that influence the optimal capital structure? (3)

- **Business risk** — volatile earnings call for less debt.
- **Tax position** — interest is tax-deductible, so the debt tax shield favours debt for tax-paying firms.
- **Asset tangibility, growth stage & interest rates** — more collateral and mature, stable operations support more debt; growth firms and high rates favour equity.

1.3 Purpose of equity in the capital markets? (3)

- **Capital raising** — channels savings into companies to fund expansion, R&D and operations.
- **Ownership & participation** — lets investors own a stake and share in growth, with price discovery.
- **Risk distribution** — spreads risk across many investors, supporting liquidity and market stability.

1.4 Role of equity in a company's capital structure? (3)

- **Permanent, risk-absorbing capital** — the cushion that bears the first losses, with no obligation to repay.
- **Funds assets & balances debt** — finances operations and is balanced against debt to reach the optimal structure.
- **Confers ownership & supports borrowing** — gives control and a residual claim; a solid equity base underpins further debt capacity.

1.5 How does the Debt-to-Equity (D/E) ratio help assess financial risk? (3)

- **Measures leverage** — D/E = total debt ÷ total equity; shows reliance on borrowing vs owners' funds.
- **Higher D/E = higher financial risk** — more fixed obligations → greater default risk and earnings volatility in a downturn.
- **Enables comparison** — compare gearing across firms and judge debt-servicing ability; a very low D/E may signal an under-used tax shield.

QUESTION 2 · EQUITY INCOME, SHARE TYPES, LISTED/UNLISTED, DRS [15]

2.1 Two primary forms of income from investing in equities? (3)

- **Dividends** — distributions of the company's profit to shareholders.
- **Capital gains** — the increase in the share's price between buying and selling.
- (For foreign shares, **FX gains/losses** can be a further source.)

2.2 How does investing in preference shares differ from ordinary shares? **(3)**

- **Income:** preference pays a **fixed** dividend; ordinary pays a **variable**, non-guaranteed dividend.
- **Priority:** preference ranks **ahead** of ordinary for dividends and on liquidation.
- **Rights & risk:** preference usually has **little/no voting** → lower risk, bond-like; ordinary carries voting and more risk/upside.

2.3 Significance of voting rights for ordinary shareholders? **(3)**

- **A say in governance** — elect directors, approve mergers/takeovers, appoint auditors.
- **Influence & control** proportional to the holding — the mechanism that holds management accountable to owners.
- **Protection** — voting (e.g. cumulative voting) lets shareholders, including minorities, defend their interests.

2.4 Main differences between listed and unlisted equity securities? **(3)**

- **Listed:** traded on a licensed exchange (public), continuous trading, regulated (FMA), transparent pricing, more liquid.
- **Unlisted:** traded OTC via dealers/market makers (private), irregular trading, lightly regulated, opaque pricing, illiquid.
- Unlisted is **riskier** — no formal market, wide spreads, possible price manipulation by a single market maker.

2.5 Role & characteristics of a depository receipt (DR) in foreign investment? **(3)**

- **What it is:** a negotiable instrument issued by a **bank** representing an interest in a foreign company, trading on a domestic exchange like a common share.
- **Role:** facilitates cross-border investment at **lower cost**, in the investor's local market and currency.
- **Characteristics: not** issued by the company (raises it no capital); typically **no voting rights** (held by the bank); known as GDRs / ADRs.

QUESTION 3 · DETERMINANTS OF OPTIMAL CAPITAL STRUCTURE [15]

3.1 How does a company's lifecycle stage influence its optimal capital structure? **(3)**

- **Start-up/growth:** uncertain cash flows & few tangible assets → rely on **equity** (e.g. venture capital), little debt.
- **Mature:** stable cash flows & assets → can carry **more debt** and use the tax shield.
- **Decline:** reduce debt / return capital; over-gearing here sharply raises distress risk.

3.2 Why is business risk a determinant of optimal capital structure? **(3)**

- **Business risk** = the variability of a firm's operating earnings.
- Adding debt adds **financial risk** on top — high business risk + high debt **compounds** total risk.
- So high-business-risk firms should use **less debt** to keep total risk (and distress probability) manageable.

3.3 Why does increasing leverage initially raise firm value, and what risk offsets this? (3)

- Debt is **cheaper** than equity and interest is **tax-deductible** — the **tax shield** lowers WACC and raises firm value at first.
- But as debt rises, the probability and cost of **financial distress / bankruptcy** increase.
- Beyond the optimal point, distress costs **outweigh** the tax shield and value falls (trade-off theory).

3.4 What role does lender risk-appetite play for early-stage companies? (3)

- Early-stage firms are **risky** — no track record, few assets, uncertain cash flows.
- Lenders are therefore **reluctant**, and demand high rates or collateral the firm lacks.
- This effectively **caps debt availability**, forcing reliance on **equity** (angel/venture capital).

3.5 How do agency costs affect capital-structure decisions? (3)

- **Agency cost of equity:** managers may waste free cash flow — debt's fixed payments impose **discipline** (less free cash to misuse).
- **Agency cost of debt:** shareholders may take excessive risk / underinvest at lenders' expense → lenders add **covenants** and charge more.
- The capital structure is set to **balance** these opposing agency costs.

QUESTION 4 · DEBT VERSUS EQUITY [10]

4.1 Primary difference between debt and equity (obligations & ownership)? (2)

- **Debt** = borrowing → contractual obligation to repay principal + interest; holder is a **lender**, not an owner.
- **Equity** = ownership → **no** repayment obligation; holder is an **owner** with a residual claim.

4.2 How do equity investors' objectives differ from debt investors'? (2)

- **Equity investors:** capital growth + dividends, accepting higher risk for higher potential return (plus voting).
- **Debt investors:** stable income (interest) + capital preservation, lower risk, priority claim.

4.3 Characteristics of preference shares vs ordinary shares? (2)

- **Preference:** fixed dividend, paid before ordinary, priority on liquidation, little/no voting → lower risk.
- **Ordinary:** variable dividend, residual claim (last), voting rights → higher risk/return.

4.4 Why are equity investors owners, and how does this affect their claims? (2)

- They hold a **share of ownership** → **residual claimants** on profits and assets.
- So they rank **last** on liquidation (after creditors & preference holders) — most risk, but unlimited upside.

4.5 Main advantage for a company issuing equity instead of debt? (2)

- **No repayment obligation / no fixed interest** — permanent capital that lowers financial risk and strengthens the balance sheet.

QUESTION 5 · PRIMARY & SECONDARY MARKETS, IPOS, LISTING [30]

5.1 Main difference between primary and secondary equity markets, and their functions? (5)

- **Primary market:** the company issues **new** shares and **raises capital** (IPO/SEO) — its function is **capital formation**.
- **Secondary market:** investors trade **existing** shares among themselves; the company receives **no new capital**.
- Secondary functions: **liquidity** (enter/exit easily), **price discovery**, and fair, continuous valuation.
- They are linked: investors buy in the primary market **because** the secondary market lets them sell later.

5.2 Explain the IPO process and how it helps a company raise capital. (5)

- Decide to go public → appoint **advisers/underwriters** → due diligence & **prospectus**.
- **Valuation & price-setting** (book-build) → regulatory/JSE approval → offer to investors → **listing & allotment** → trading begins.
- It raises a large pool of **permanent equity capital**, widens the investor base, and gives the firm **shares as currency** for future deals plus a higher profile.

5.3 Advantages and disadvantages of listing on the JSE? (5)

Advantages: access to capital · liquidity for shareholders · profile & credibility · shares as acquisition currency · easier future fundraising · a transparent market valuation.

Disadvantages: listing & ongoing compliance costs · heavy disclosure/regulatory burden · some loss of control · short-term market pressure · exposure to hostile takeover · constant public scrutiny.

5.4 Difference between an IPO and a seasoned equity offering (SEO), and the effect of each? (5)

- **IPO** = a company's **first** public issue (needs a price-setting process); it turns a private firm public and raises a large first round of capital — founders may dilute or partly sell out.
- **SEO** = an **already-listed** firm issues more shares; **no price-setting** needed (a market price exists).
- An SEO raises further capital but **dilutes** existing shareholders unless they follow their rights, and can send a (sometimes negative) signal to the market.

5.5 Role of underwriters in listing equity, and the risks they face? **(5)**

- **Role:** underwriters (investment banks) **advise, value, price, market** the issue and **guarantee** it — in a firm-commitment they agree to buy any unsold shares so the company raises its target.
- **Risk:** if the issue is **mispriced or undersubscribed**, the underwriter is left holding shares and may sell at a **loss**.
- They also carry **market risk** between pricing and sale, and **reputational risk** from a failed listing.

5.6 Advantages of secondary markets for investors and companies, and their key functions? **(5)**

- **For investors: liquidity** (easy entry/exit), **price discovery**, fair valuation, and the ability to diversify.
- **For companies:** a liquid, fairly-priced share supports a **higher valuation** and a **lower cost of capital**, and makes future fundraising easier.
- **Key functions:** liquidity, price discovery, efficient capital allocation, and risk transfer.

QUESTION 6 · LISTING REASONS, METHODS, RIGHTS ISSUE, RISKS [15]

GIVEN The JSE offers secure primary & secondary markets; **Company ABC** plans to list on the JSE.

6.1 Reasons for seeking a stock-market listing? **(6)**

- **Raise capital** for growth and expansion.
- Provide **liquidity / an exit** for existing shareholders.
- Raise the firm's **profile and credibility**.
- Obtain a transparent **market valuation**.
- Use **shares as acquisition currency** and ease future fundraising.
- Broaden the **investor base** and help attract/retain staff (share schemes).

6.2 Describe two methods of obtaining a listing. **(2)**

- **Initial Public Offering (offer for subscription/sale):** the company offers shares to the public and raises capital as it lists.
- **Introduction:** already-issued shares are listed **without** a new offer/raising capital (used when the shareholder base is already wide).

6.3 A listed company raises R2bn in equity (rights issue). If you do NOT follow the rights issue, what happens to your holding? (3)

- Your shareholding is **diluted** — your **percentage ownership and voting power fall** as new shares are issued.
- The share price typically drops to a **theoretical ex-rights price**, reducing the value of your existing holding.
- You can **sell your nil-paid rights** to partly offset the dilution; if you do nothing, you forfeit that value too.

6.4 Risks of trading in the equity market? (4)

- **Market/price risk** — volatility can cause capital loss (price can fall below cost).
- **Liquidity risk** — you may not sell quickly at a fair price.
- **Company-specific (business) risk** — poor performance hurts the share; dividends aren't guaranteed.
- **Systematic / economic risk** (and currency risk on foreign exposure).

QUESTION 7 · GORDON GROWTH MODEL [5]

GIVEN Blue-chip telecoms: dividend = **820c**, growth **g = 1%**, required return **r = 8%**.

7.1 Shortcomings of the Gordon Growth model to keep in mind. (5)

Context (the model's output): Value = $D_0(1+g)/(r-g)$ = 820 × 1.01 ÷ (0.08 − 0.01) = 828.2 ÷ 0.07 ≈ **11 831c ≈ R118.31**. Its shortcomings:

- Assumes a **constant, perpetual growth rate** — unrealistic for real companies.
- **Highly sensitive to inputs** — a small change in r or g causes a large change in value.
- **Breaks down if g ≥ r** (gives a negative or infinite value).
- Only suits **mature, stable dividend payers** — not high-growth or non-dividend firms.
- Ignores other value drivers (qualitative factors, changing risk) and relies on **estimated** inputs.

QUESTION 8 · MARKET CAPITALISATION — FSR VS NEDBANK [10]

GIVEN FirstRand (FSR) share price **R47.33**; Nedbank (NED) share price **R206.50**. Does NED's higher price mean it is more valuable?

8.1 Describe market capitalisation using these two companies. (4)

- **Market capitalisation = current share price × number of shares in issue.**
- It measures the **total market value of a company's equity** — not the share price on its own.
- R47.33 and R206.50 are only **per-share** prices; to compare value you must multiply each by its shares outstanding.
- A company with a **lower price but far more shares** can have a much larger market cap.

8.2 Which of the two has the greater equity value? (2)

- You **cannot tell from the share price alone** — the company with the higher **market capitalisation** has the greater equity value.
- In practice FirstRand has **many more shares in issue**, so its market cap exceeds Nedbank's **despite** its lower share price — a higher price ≠ more valuable.

8.3 Describe four benefits of being a listed company. (4)

- **Access to capital** for growth.
- **Liquidity / marketability** of shares for owners.
- **Profile, credibility** and a transparent market valuation.
- **Shares as acquisition currency** and easier future fundraising.

QUESTION 9 · VALUATION TECHNIQUES BY SECTOR [10]

9.1 Give an example of a listed company in that (telecommunications) sector. (1)

MTN Group (or Vodacom Group / Telkom) — all listed on the JSE.

9.2 List three valuation techniques, the companies/sectors each suits, and why. (9)

TECHNIQUE	SUITED TO	REASON
Dividend Discount Model (DDM / Gordon Growth)	Mature, stable dividend payers — telecoms, banks, utilities	Regular, predictable dividends make the PV-of-dividends meaningful.
Discounted Cash Flow (FCFF / FCFE)	Firms with no/irregular dividends but predictable cash flows — industrials, growth firms	Values the intrinsic cash-generating ability directly, even without dividends.
Relative valuation (P/E, EV/EBITDA)	Firms with comparable listed peers and positive earnings — retail, consumer	Quick, market-based benchmark against similar companies.

(An **asset-based / NAV** model also suits asset-heavy firms — property/REITs, investment holding companies — where value = assets − liabilities.)

QUESTION 10 · DUAL-CLASS SHAREHOLDING CASE STUDY [25]

GIVEN **Naspers** (N shares = 1 vote; unlisted A shares = 1 000 votes, held by management) and **Alphabet/Meta** (multi-class: high-vote insider shares + low/no-vote public shares) — dual-class structures that concentrate control with founders/insiders.

10.1 How do these structures impact corporate governance? (3)

- They **concentrate control** with insiders and breach the "**one share, one vote**" principle.
- They **weaken board accountability** to ordinary shareholders and can **entrench** management.
- Trade-off: greater **stability & long-term vision** vs weaker checks and balances.

10.2 Could concentrated insider voting compromise accountability to shareholders? (3)

- **Yes** — high-vote holders can **override** ordinary shareholders and resist shareholder activism.
- Ordinary (N / Class A) holders can **never outvote** the insiders, however many shares they own.
- Management can act with **fewer checks**, reducing accountability.

10.3 How might these structures affect an investor's decision to buy? (3)

- Some investors **avoid** them, seeing a **governance risk** and limited say.
- Others **accept** them where management has a strong track record (e.g. Naspers → Tencent).
- The decision turns on **trust in management** and the **price/valuation** offered.

10.4 Do investors perceive higher risk from diminished voting power, and how is it reflected in price/valuation? (3)

- **Yes** — weaker voting rights are seen as a **governance risk**.
- This often produces a "**governance discount**": low/no-vote shares (e.g. Naspers N, Alphabet Class C) tend to trade at a **discount** to high-vote shares.
- Investors demand a **higher required return** → a **lower valuation**.

10.5 Would the companies make different strategic decisions if voting power were more equal? (2)

- Likely **more short-term, shareholder-pressured** decisions and **fewer bold long-term bets**.
- Transformational moves (e.g. Naspers's early Tencent investment) might **not** have happened under short-term pressure.

10.6 How do these structures serve as a defence against hostile takeovers? (2)

- Insiders' concentrated votes mean an acquirer **cannot gain control** simply by buying public shares.
- The structure therefore acts as an effective **takeover shield**.

10.7 Other mechanisms to maintain control without multi-class shares? (2)

- **Poison pills**, **staggered/classified boards**, and **supermajority** provisions.
- **Golden shares**, **shareholder agreements**, or a **pyramid/holding-company** structure.

10.8 How do these structures affect market perception of transparency & governance? (2)

- They can **signal weaker governance/transparency**, a reputational concern; some indices exclude dual-class firms.
- But **strong performance** can offset the concern and keep investors comfortable.

10.9 How sustainable are these structures in the long run? (3)

- They can **persist while founders deliver** and continue to hold the high-vote shares.
- **Pressure builds** over time — activism, index exclusion, and succession (founder exit) test them.
- **Sunset clauses** (auto-converting to one-share-one-vote) are increasingly used; structures may be unsustainable if performance lags.

10.10 Could these companies alter their structures in future? (2)

- **Yes** — triggers include founder exit/death or dilution below a threshold, investor/regulator/index pressure, or a sunset clause converting to one-share-one-vote.
- Large **capital needs** requiring broader equity could also force a change.

SUBMISSION REMINDER Write your final answers in your **own words**, number them to match the workbook, and put your name, surname & ID on every page. Pass mark = **65%**.